

Sleep Project Final Report

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1. Introduction

Sleep is a critical yet understudied health behavior that influences physical health and well-being. Importantly, research has identified sleep health, which includes multiple sleep parameters above and beyond sleep duration (e.g., quality, consistency, etc.) as a critical intervention target to improve overall public health and target inequities.¹⁻⁴ Adolescence is a particularly important developmental period in which to address sleep health. Due to a “perfect storm” of biological, psychosocial, and societal influences, good sleep health is rarely achieved during adolescence,^{5,16} and adolescents struggle to meet the recommended 8-10 hours^{17,18} of sleep duration required to support healthy cognitive/attention and emotional functioning.^{5,16} Additionally, inconsistent sleep timing, particularly between weekdays and weekends, has been associated with poor mental health outcomes and increased BMI.^{19,20} Not only do adolescents struggle to maintain healthy sleep, research has shown that poor sleep in adolescence is associated with continued poor sleep and other mental and physical health consequences in adulthood,²¹⁻²³ making this period an important window of opportunity for intervention.

Disparities exist in sleep health, with minority youth experiencing worse sleep health compared to White non-Hispanic peers,⁶ which is proposed to stem from the same racism-related factors that contribute to other health inequities.^{3,4} One mechanism that has been proposed to explain sleep disparities is stress.^{1,3} Stress can contribute to poor mental health and increased arousal and hypervigilance both of which have been shown to impact on sleep health.¹¹ Discrimination, a unique stressor that induces feelings of threat, has been associated with worse sleep quality among adolescents even after controlling for other forms of stress.⁹ However, there may be protective or promotive factors at various levels of influence including

individual (e.g., resilience/coping, emotion regulation, substance use, screen time, physical activity, sleep hygiene), family (e.g., sleep environment, family conflict and cohesion, chaos, routines, functioning, resilience), and neighborhood (e.g., safety, community cohesion, light, noise, greenspace) levels that can alter the relations between stress experiences and sleep health. For example, research has demonstrated that children in a low socioeconomic status home with a high quality home environment (e.g., calm, parental involvement/responsivity) exhibit better sleep health.¹² In addition, adolescents residing in homes where parents set bedtimes and with better family functioning report better sleep health.¹⁴ Further, adolescents residing in neighborhoods with higher levels of cohesion experience fewer sleep problems.¹⁵ However, research has not examined protective factors at multiple levels to understand how to most effectively intervene on sleep health.

Drawing on a socioecological model of pediatric sleep health (see Appendix A)⁴ and responding to recently developed research priorities in sleep disparities,¹ the present study seeks to use multiple methods (i.e., surveys, diaries, actigraphs, geocoding) to delineate factors that directly contribute to and/or mitigate experiences of stress and discrimination to promote sleep health in a sample of urban adolescents in Chicago with the ultimate goal of creating a resilience-based sleep health intervention to foster health equity.

Inadequate sleep has widespread effects on health and well-being. Given that sleep is a modifiable health behavior, interventions that target sleep health (e.g., duration, quality, consistency) offer promise for addressing health inequities. Adolescence is a time of inadequate sleep,⁵ and minority youth report less time, poorer quality, and less consistent sleep than their White peers.⁶ Identifying key contributing factors to sleep health among urban adolescents is essential to delivering effective interventions.⁷ One proposed contributor to these health-related disparities is the stress that minority youth experience due to discrimination, racism, and resulting social and structural inequities.⁸⁻¹¹ Stress, including daily experiences of discrimination, has been shown to influence sleep, but this has been examined primarily among minority youth residing in small towns and rural settings. Protective factors at various levels of influence including individual, family, and neighborhood may buffer the effect of these stress experiences on sleep health.^{7,12-15} However, research has not examined protective factors at multiple levels simultaneously to understand how to most effectively intervene on sleep health. The current study will be the first to delineate which factors at the individual, family, and neighborhood levels make important collective contributions to sleep health among a sample of urban adolescents.

2. Methods

For this project, we used PUSH adolescent daily diary, and actigraphy data. This data recorded various metrics related to sleep, and can be found in the appendix. The diary data came from two separate files that were hand recorded. To ensure that the data was as accurate as possible, we had two separate researchers transcribe data, and then we compared the data to account for possible errors. The actigraphy data was collected from a wearable band by the

subjects of the study, and was merged with diary data for each participant. Even after cleaning the diary data, we still had issues with interpreting some of the results from the subjective data, and in the end, we were left with 172 participant nights from 28 participants that were able to be used for analysis.

The two separate diaries that were reconciled contained many missing value codes that we converted to NA. We standardized column names after gaining access to the data sets and used one cleaned diary file as the base, filling in missing values from the second clean file when available. After the handwritten diaries were combined, we then combined it with online diary data in the same sleep study to have one overarching diary data set, which we can then merge with the actigraphy data set. This way, we ensure that everything is combined correctly and accurately, so that we can evaluate sleep profiles completely.

In the merge data set, the date and clock fields arrived in various mixed formats, such as Excel serial dates, text dates, and POSIX datetimes. We converted all of these types to one standard date format. For time fields in the data set that were of a different format, if the clock and decimal versions agreed within five minutes, the clock entry was used. If there was a decimal value that looked like clock notation, such as 7.45, it was treated as clock notation, especially when 7:45 was a time recorded for that same person in a different diary data set. If two versions of a time entry differed by 12 hours, the discrepancy was treated as an AM/PM flip. If a time entry could not be cleanly reconciled, we flagged it for manual review and removed it from an analysis ready data set until the data could be manually verified.

We merged datasets by participant IDs into a common PUSH_### format. We then converted study day labels such as “Day 3” to numeric day values. We merged diary and actigraphy data by the normalized participant ID and study day. We did this instead of merging by date alone, since diary dates and actigraphy dates sometimes differed by one calendar day, which we attribute to manual transcription error. After merging, we were left with 180 matched diary actigraphy rows, 125 diary only rows, and 24 actigraphy only rows. For this study, we only considered matched rows for the main analysis. For the main analysis data set, we kept 172 matched rows, where rows were kept if dates matched exactly, differed by only one day, had a missing diary date but valid actigraphy date, or had an obvious year typo that could be edited. More specifically, 61 rows had matching raw dates, 87 rows had a one day diary actigraphy difference, 16 rows had a missing diary date but valid actigraphy date, and 8 rows had an obvious year typo and were repaired. Of the 180 rows, we kept only 172, since eight matched rows had date gaps that were too large to confidently treat as the same sleep episode. We emphasize that diary only and actigraphy only rows are not necessarily bad, they were just excluded from paired analysis because they lacked the information necessary for diary actigraphy modeling.

We created several variables for our data set. The first is Screen Time total, which combines hours and minutes for things like television, videos, video chat, social media, and more. We specified the variable as such:

$$\text{Total Minutes}_{ij} = 60 \times \text{Hours}_{ij} + \text{Minutes}_{ij}$$

We also created weekend night indicators for Friday and Saturday night and summer break indicators using Chicago public school school calendar windows. For sleep timing variables, sleep onset and offset were represented as decimal hours, and to account for this, we used a circular encoding of time because it wraps cleanly around midnight. The variables are specified as such:

$$\theta_{ij} = \frac{2\pi t_{ij}}{24}, \quad s_{ij} = \sin(\theta_{ij}), \quad c_{ij} = \cos(\theta_{ij})$$

$$\bar{t} = \frac{24}{2\pi} [\text{atan2}(\bar{s}, \bar{c}) \bmod 2\pi]$$

We also defined the categories of “good sleeper” and “bad sleeper”. Daily top quartile sleep nights were defined using self-reported sleep quality, with 75th percentile cut off was 9, so nights with sleep quality at least 9 were defined as top quartile nights. Participant level good sleepers were defined by each participant’s mean sleep quality, where the participant-level 75th percentile cutoff was about 8.08:

$$\bar{Q}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} Q_{ij}$$

$$\text{GoodSleeper}_j = \mathbf{1}(\bar{Q}_j \geq Q_{0.75}(\bar{Q}))$$

We were interested in linear regression and mixed effects modeling for analysis of our data set. The mixed affects model is key because repeated nights are nested within participants, meaning that observations from the same adolescent are not necessarily independent. Therefore, the mixed effects model addresses this by giving each participant their own personal random intercept. We calculated the ICC to be about 0.13, so some clustering by participant existed.

The main mixed effects model predicted sleep quality from sleep duration, sleep, onset, weekend, and summer break, and the model we used is the following:

$$\text{DurationWithin}_{ij} = \text{Duration}_{ij} - \text{Duration}_j^{\bar{}}$$

$$Q_{ij} = \beta_0 + \beta_1 \text{DurationWithin}_{ij} + \beta_2 \text{Duration}_j^{\bar{}}$$

$$+ \beta_3 \text{OnsetWithin}_{ij} + \beta_4 \text{Onset}_j^{\bar{}} + \beta_5 \text{Weekend}_{ij} + \beta_6 \text{Summer}_{ij} + b_{0j} + \varepsilon_{ij}$$

The model used complete cases for the variables included in that model. Here, we used 147 nights from 27 participants. It is also important to note that all analysis should be interpreted as association, and not causal effect. We also note that good sleeper status is an empirical classification within the sample, and is not from a clinical diagnosis.

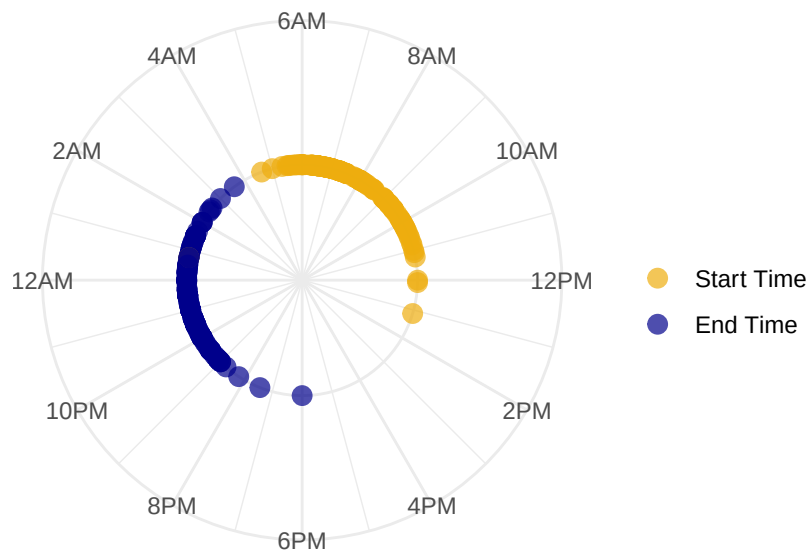
3. Results

Descriptive Results

Descriptive results: Through our upper quartile study on sleep quality we discovered that the top-quartile sleepers had a much longer sleep duration (*275 minutes higher*) as well as lower caffeine (*20% less*) use and less device time (*12 minutes less*) compared to sleepers who were not in the top-quartile. Unexpectedly, top sleepers also had higher anxiety ratings (*1.06 points higher*), more social media use(*18 minutes higher*), and less physical activity(*11.50 minutes less*).

The average overall wake time was reported to be around 7:30 AM (*7.665845*) and the average sleep time was around 11:00 PM (*23.26906*).

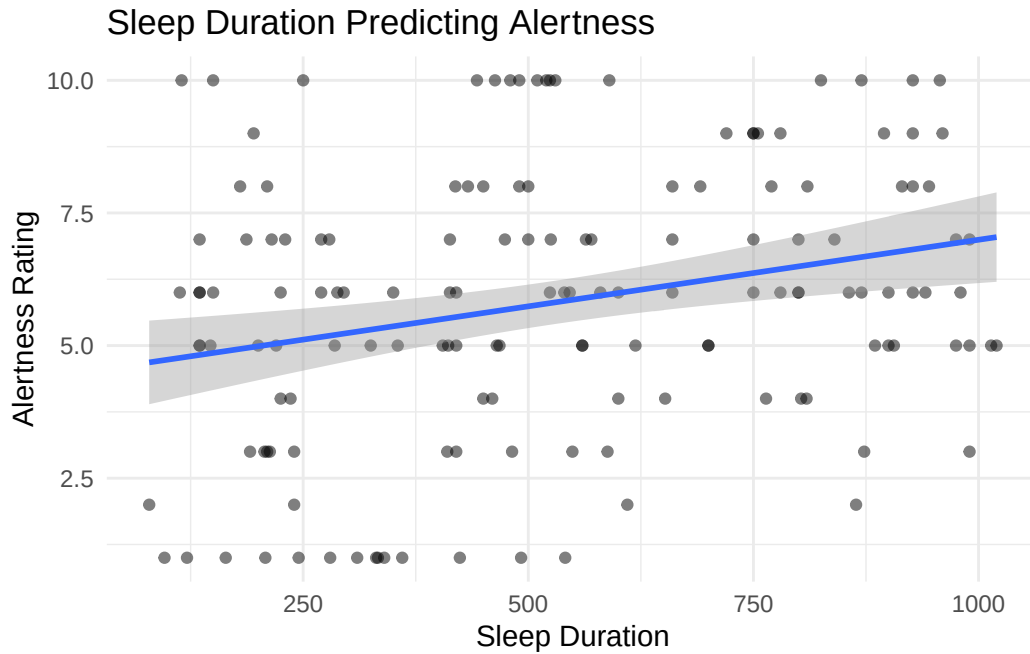
Sleep Start and End Times (Clock View)



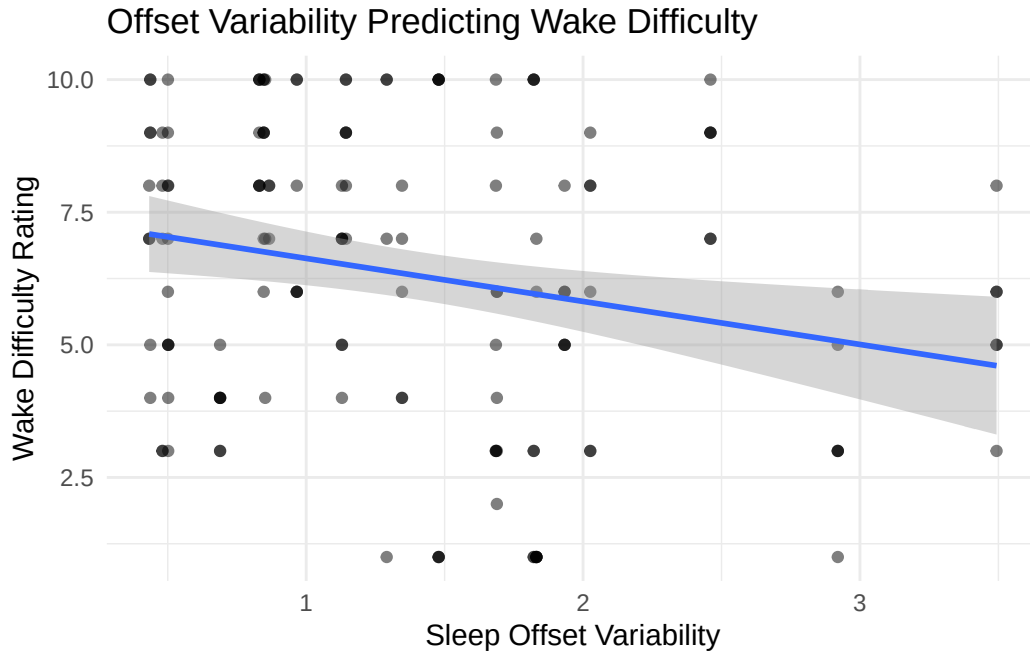
Regression Results

The regression study was based on the independent variables: onset time, offset time, onset variability, offset variability, sleep duration and the dependent variables: alertness, anxiety,

wake difficulty, and sleep quality. Due to the recurring problem of heteroscedasticity violating the assumption of homoscedasticity in the linear regression analyses conducted for each dependent variable we used an HC3 heteroskedasticity-robust standard errors test to interpret coefficients more accurately. The results showed that sleep duration is a significant predictor for alertness ($p=0.0096$, $\beta=0.0023$), indicating that longer sleep duration is associated with higher alertness. This made worked with our assumption that a longer/better sleep leads to a stronger sense of alertness throughout the day.



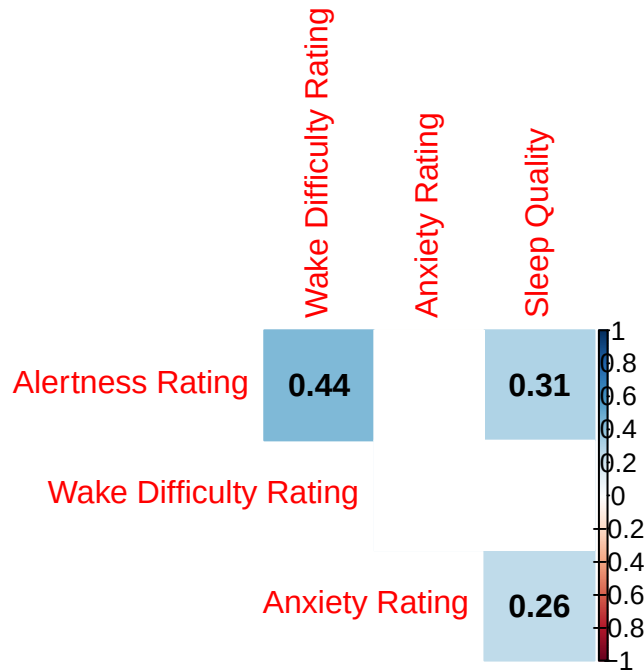
Another notable result is offset variability as a significant predictor for wake difficulty ($p=0.006$, $\beta=-0.720$), indicating that a greater inconsistency in wake time is associated with decreased difficulty. This was particularly interesting as it worked against our assumption that a more consistent wake time would lead to less difficult wake process aka lower wake difficulty ratings.



There were no other significant results for the remaining dependent variables: anxiety and sleep quality. However this is not to say that there are no predictors for these variables but just that there is further analysis that can be done to get a better grasp at what indicates anxiety and sleep quality reports from adolescents.

Correlation Results

We also looked into correlation between the dependent variables and found moderate correlations between alertness & wake difficulty ($r=0.42$), alertness & sleep quality ($r=0.31$), and anxiety & sleep quality ($r=0.30$).



All of the reported correlations are positive, showing us that the variables tend to increase together. The relationships are meaningful however they are not strong showing a range of around 0.3-0.4 for r . Higher alertness is associated with both easier wake times and better sleep quality which matches up with our assumptions on the effects of sleep quality on alertness. However, it also adds the additional perspective that an easier reported wake time is associated with higher alertness, something we didn't think to look into prior to this analysis. The correlation that anxiety has with sleep suggests that a higher anxiety is linked to a worse perceived sleep quality, this makes sense as someone who isn't as well rested may report a stronger sense of anxiety due to the lack of rest. These results indicate the dependent variables are related but not redundant meaning they capture overlapping but distinct aspects of the adolescent sleep experience within our data. Overall, the results support the idea that subjective sleep experiences/ratings are interconnected.

Mixed Effects Results

The data included 147 nights from 27 participants, within the data we noticed some trends although not statistically significance still peaked our interest. The trends we found were the association of more sleep than usual to a better reported sleep quality as well as a later sleep onset associated with a worse reported sleep quality. Though these are interesting trends it's important to emphasize that no predictors reached statistical significance in this study. The model assumptions were mostly satisfied showing no major variance issues and although there were some slight left-skew residuals the overall model fit is considered acceptable.

Takeaways

Sleep duration is the strongest consistent predictor within our study. However, most other sleep variables showed limited predictive powers. Some relationships exist between key outcomes, but casual predictors are weak overall.

4. Conclusions/Future Work

In this project we worked as a data consulting team to clean and merge adolescent sleep data from the PUSH study, with some analysis at the end. Our work involved resolving discrepancies between multiple datasets. Mainly, handling missing values and outliers as well as integrating sleep diary data with actigraphy measures to create an analysis-ready dataset. After preparing the data, we conducted descriptive analyses, regression models, correlation tests, and mixed effects modeling to explore relationships between sleep patterns and outcomes such as alertness, anxiety, wake difficulty, and overall sleep quality.

The results showed some noticeable patterns. Descriptively, adolescents in the top quartile of sleep tended to have longer sleep duration and healthier behavioral patterns, such as lower caffeine use and reduced device time. Surprisingly, some unexpected trends such as higher anxiety and social media use were also observed. When we ran our regression analyses, we found that sleep duration was a significant predictor of alertness. This indicates that longer sleep is associated with improved alertness the next day. We also found that sleep offset variability significantly predicted wake difficulty. Despite these findings many predictors were not statistically significant across models. Mixed effects modeling showed that while the direction of relationships has some backing, these effects were not strong enough to reach significance in this sample. Finally, the correlation analyses supported moderate relationships between alertness, wake difficulty, and sleep quality.

Had we have had more time, we would encourage the researchers to expand the sample size and duration of data collection. This would greatly boost the statistical power and better capture the variability in sleep patterns. This of course is not an easy task, as adolescents are not always taking surveys with the most serious intentions. In terms of analyses, we have a strong set of techniques. However, these techniques are only as good as the data is to analyze, further pushing the need for better data.

5. Appendix

All code can be found at the following GitHub repository: https://github.com/purplebamboo2/Sleep_370E

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